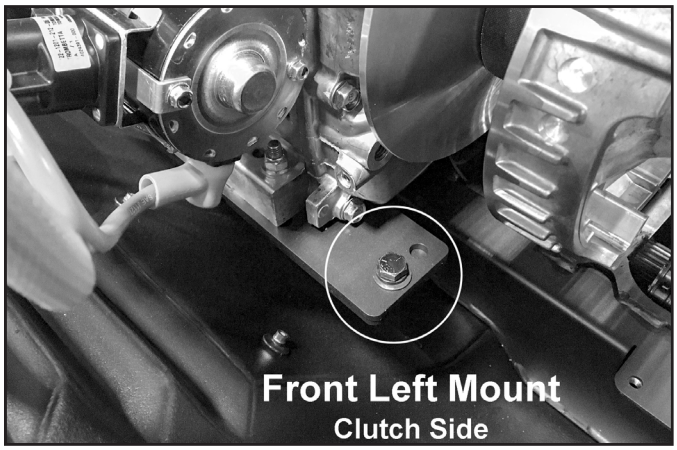
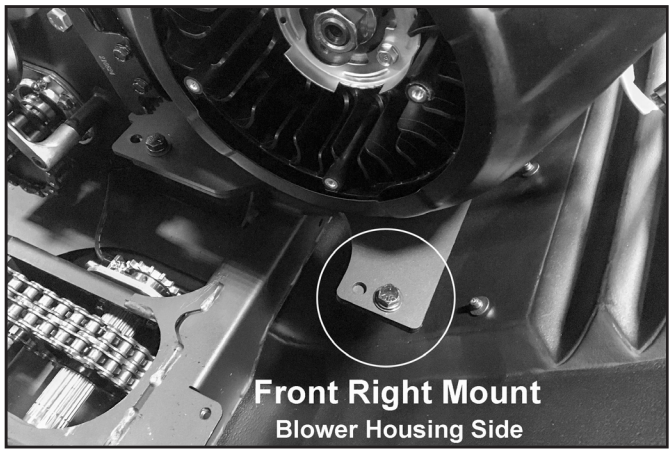
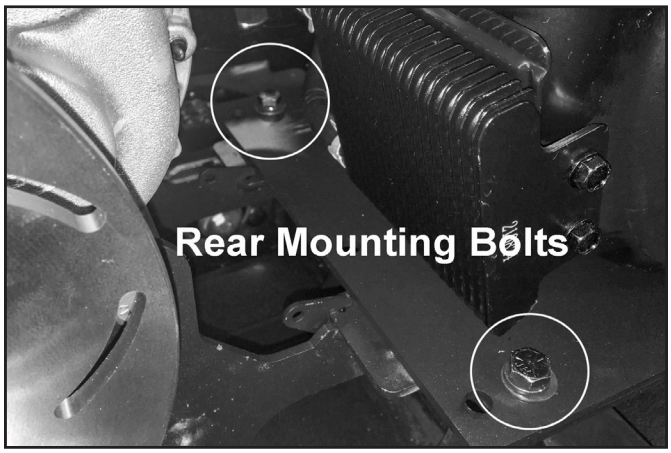
13. Remove all four (4) mounting bolts using a 9/16" socket and ratchet.







Lifting The Engine From The Vehicle

CAUTION

This procedure requires the use of an overhead hoist. Do NOT attempt to lift out by hand. Ensure all components attached to the engine have been detached. This may include any accessories that may be installed to the vehicle. Lift out slowly and carefully. If you wish to remove the muffler from the engine before raising it out, you may do so as an optional procedure. You may find it easier to maneuver the engine up and out with the muffler removed, but it is not necessary. Protect the upper body with heavy blankets to avoid any damage to the plastic!

WARNING

Ensure all hoist chains or belts are secure, and hooks are locked into place. Keep all fingers and hands clear of the area from which the engine is being lifted.

13. Attach the overhead hoist to the engine. There are lift hooks provided on each engine.

14. Slowly and gently raise the engine through the hood area with the overhead hoist. Watch for any wiring they may have been overlooked in disconnecting which may include some optional accessory kit wiring. Some manoeuvring is required as the engine does not necessarily come straight up and out. Avoid damage to any other components as well as the upper body.
15. Place the engine on a clean work bench for service required.
16. Remove the engine from the engine mounting plate.




Installing The Briggs 650/700 Engine

1. Install the engine mount plate to the engine block securing with four (4) HWHCS 5/16-18x2.00 GR8 YZ and four (4) HXFNN, 5/16-18 GR8 YZN. **Torque to 295 in-lbs (dry, non-lubricated condition).**
2. Slowly and gently lower the engine through the hood area. Some maneuvering is required as the engine does not necessarily drop straight in. Avoid damage to any vehicle components as well as the upper body.
3. Align the engine mounted frame holes with the holes in the lower main frame and reinstall each of the four (4) HHCS 3/8-16x2.0 GR5 BNZ & 108-06 Lock washers. Secure and torque each fastener to **31 FT-LBS** (Dry, Non-Lubricated Condition)
4. Reconnect all engine sensor wiring between engine and main vehicle wire harness (700 models only).
5. Reconnect engine connector plugs in the main wire harness located at the front of the engine.
6. Reconnect fuel vent from fitting on engine.
7. Reconnect the intake hose from the air filter cannister.
8. Reconnect the throttle cable at the engine cable retainer.
9. Reconnect the ground cable at the engine.
10. Reconnect the fuel line and return fuel line from the engine.

11. Locate the tailpipe assembly and reattach to the muffler with connecting springs.
12. Reinstall the heat deflector shield over the muffler.
13. Reconnect the battery positive (+) and negative (-) cables.

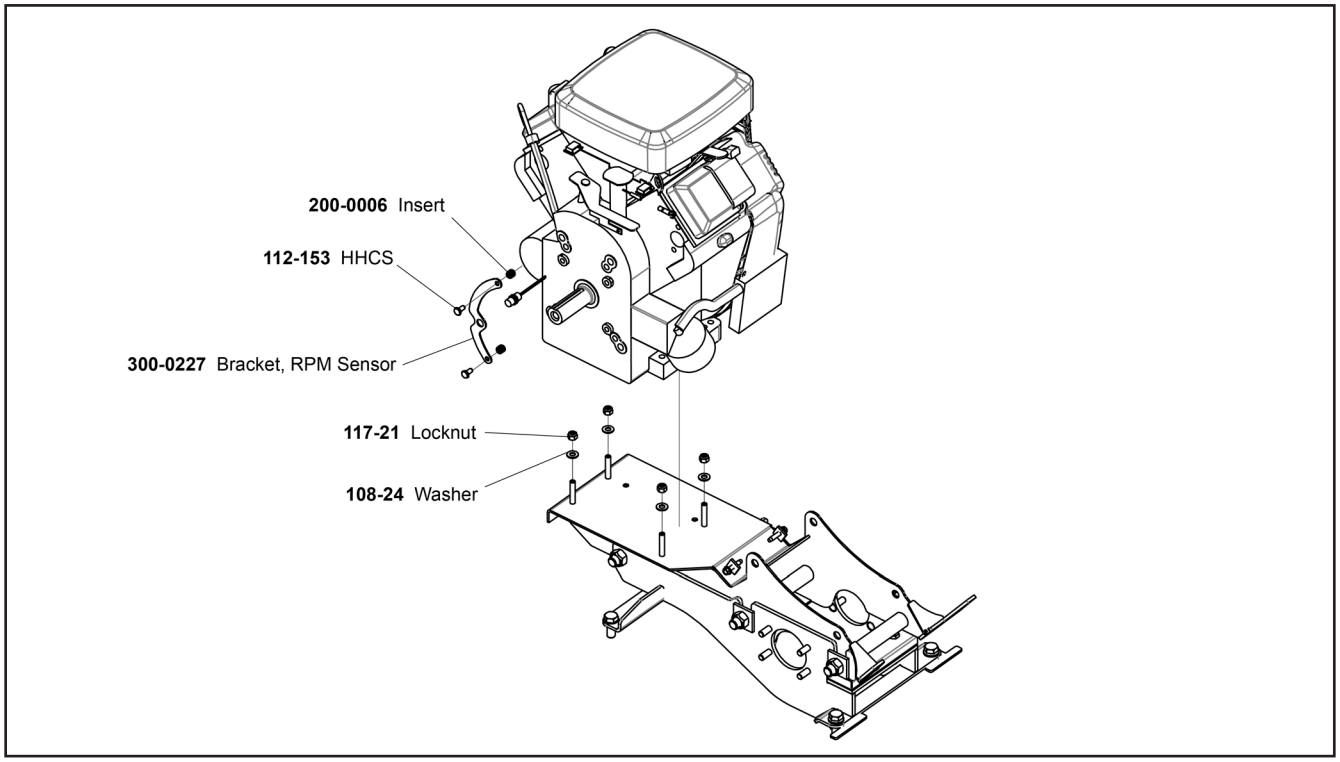
IMPORTANT

Reconnect the Positive (+) cable first, followed by the negative cable.

IMPORTANT

Refer to the diagram for detailed information on the engine mounting hardware.

Removing The Briggs 600 Engine From The Vehicle.

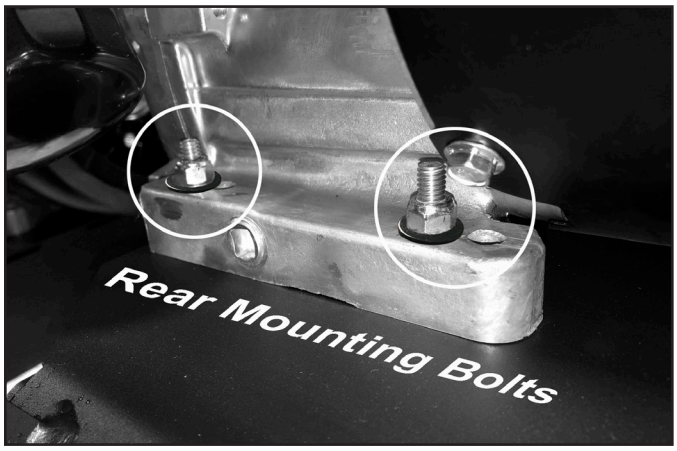


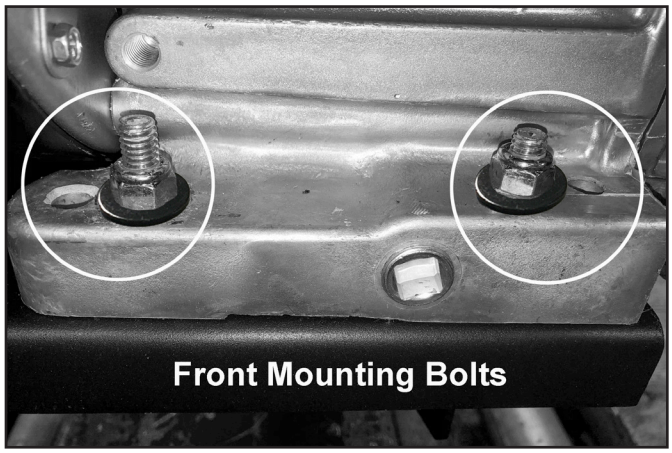
1. Disconnect the battery positive (+) and negative (-) cables from the battery.
2. Remove the heat deflector shield covering the muffler.
3. Locate the tailpipe assembly and detach the connecting springs between the tailpipe assembly and muffler. Pull the tail pipe assembly out through the hood area.
4. Disconnect the fuel line and return fuel line from the engine.
5. Disconnect the ground cable at the engine.
6. Disconnect the throttle cable at the engine cable retainer.
7. Disconnect fuel vent from fitting on engine.

8. Disconnect engine connector plugs in the main wire harness located at the front of the engine.
9. Disconnect the wiring at the starter motor.
10. There are only four (4) nuts securing the engine to the power pack frame front & rear.
11. Remove all four (4) mounting nuts & flat washers using a 9/16" socket and ratchet.

CAUTION

There are 4 mounting holes provided in the engine block. Pay attention to which holes the powerpack frame studs are aligned with when reinstalling the engine. Mounting to the wrong holes will cause problems with clutch offset and drive belt alignment.





Lifting The Engine From The Vehicle

CAUTION

This procedure requires the use of an overhead hoist. Do NOT attempt to lift out by hand. Ensure all components attached to the engine have been detached. This may include any accessories that may be installed to the vehicle. Lift out slowly and carefully. If you wish to remove the muffler from the engine before raising it out, you may do so as an optional procedure. You may find it easier to maneuver the engine up and out with the muffler removed, but it is not necessary. Protect the upper body with heavy blankets to avoid any damage to the plastic!

WARNING

Ensure all hoist chains or belts are secure, and hooks are locked into place. Keep all fingers and hands clear of the area from which the engine is being lifted.

12. Attach the overhead hoist to the engine. There are lift hooks provided on each engine.
13. Slowly and gently raise the engine through the hood area with the overhead hoist. Watch for any wiring they may have been overlooked in disconnecting which may in-

clude some optional accessory kit wiring. Some manoeuvring is required as the engine does not necessarily come straight up and out. Avoid damage to any other components as well as the upper body.

14. Place the engine on a clean work bench for service required.





Installing The Briggs 600 Engine

1. Slowly and gently lower the engine through the hood area. Some maneuvering is required as the engine does not necessarily drop straight in. Avoid damage to any vehicle components as well as the upper body.
2. Align the holes in the engine block with the press fit studs in the powerpack frame. Install the flat washers and secure and torque each nut to **209 in-lbs** (Dry, Non-Lubricated Condition)

CAUTION

Ensure the studs in the power pack frame are aligned with the correct holes on the engine block. Not observing this could result in belt misalignment and damage to drive belt and clutches. Lower the engine with caution or damage could result to the studs in the power pack frame.

3. Reconnect engine connector plugs in the main wire harness located at the front of the engine.
4. Reconnect fuel vent from fitting on engine.
5. Reconnect the intake hose from the air filter cannister.
6. Reconnect the throttle cable at the engine cable retainer.
7. Reconnect the ground cable at the engine.
8. Reconnect the fuel line and return fuel line from the engine.
9. Reconnect the wiring to the starter motor
10. Locate the tailpipe assembly and reattach to the muffler with connecting springs.

11. Reinstall the heat deflector shield over the muffler.
12. Reconnect the battery positive (+) and negative (-) cables.

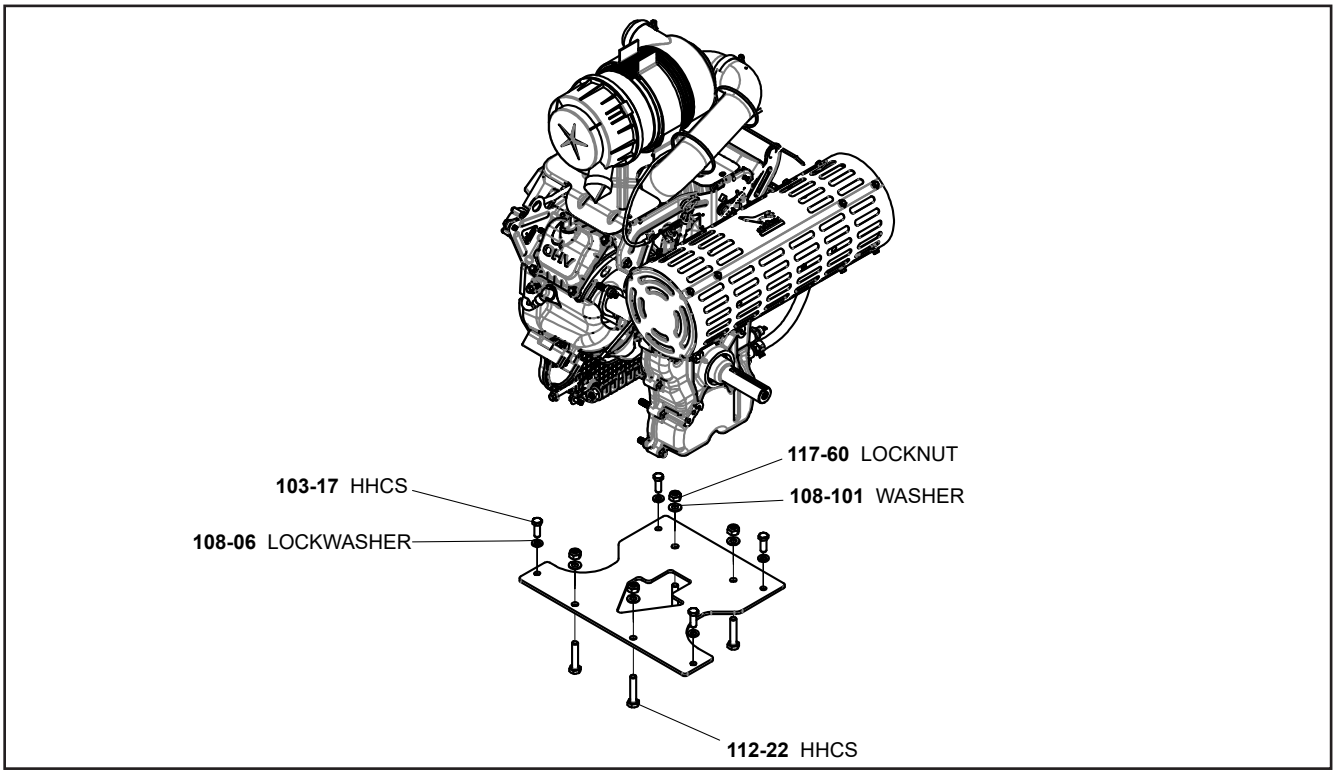
IMPORTANT

Reconnect the Positive (+) cable first, followed by the negative cable.

IMPORTANT

Refer to the diagram for detailed information on the engine mounting hardware.

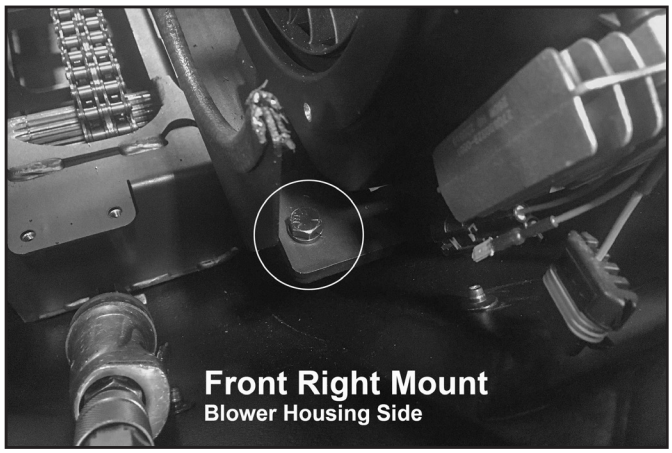
Removing The ODG 650/850 LX Engine From The Vehicle.

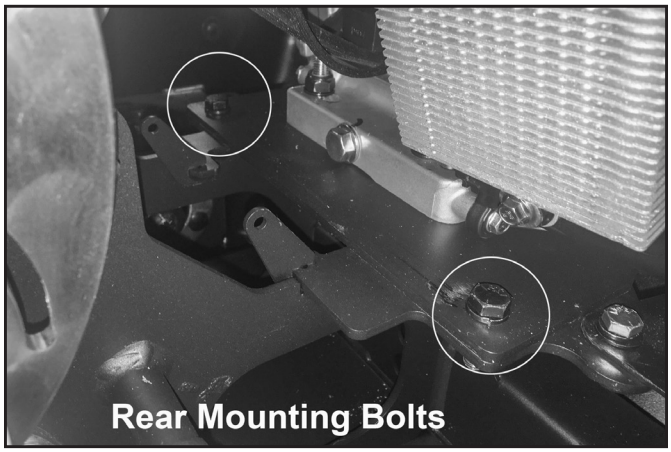


1. Disconnect the battery positive (+) and negative (-) cables from the battery.
2. Remove the heat deflector shield covering the muffler.
3. Locate the tailpipe assembly and detach the connecting springs between the tailpipe assembly and muffler. Pull the tail pipe assembly out through the hood area.
4. Disconnect the fuel line.
5. Disconnect the ground cable at the engine.
6. Disconnect the throttle cable at the engine cable retainer.
8. Disconnect fuel vent from fitting on engine.
9. Disconnect engine connector plugs in the main wire harness located at the front of the engine.

10. Disconnect the wiring at the starter motor.
11. There are only four (4) bolts & lock washers securing the engine to the main frame front & rear. **Take note of which holes in the engine mounting frame are used for installation later.**
12. Remove all four (4) mounting bolts using a 9/16" socket and ratchet.







Lifting The Engine From The Vehicle

CAUTION

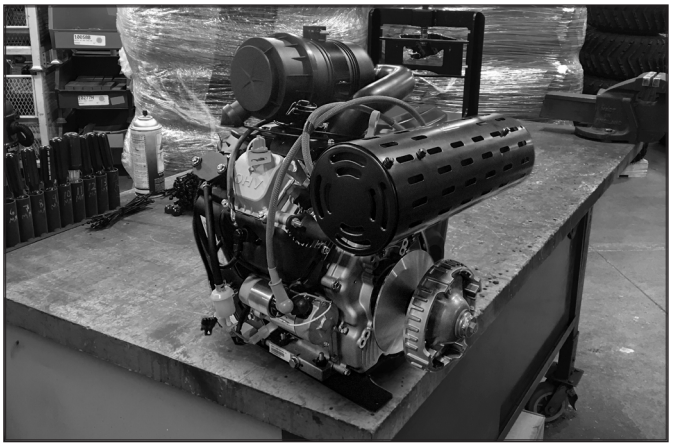
This procedure requires the use of an overhead hoist. Do NOT attempt to lift out by hand. Ensure all components attached to the engine have been detached. This may include any accessories that may be installed to the vehicle. Lift out slowly and carefully. If you wish to remove the muffler from the engine before raising it out, you may do so as an optional procedure. You may find it easier to maneuver the engine up and out with the muffler removed, but it is not necessary. Protect the upper body with heavy blankets to avoid any damage to the plastic!

WARNING

Ensure all hoist chains or belts are secure, and hooks are locked into place. Keep all fingers and hands clear of the area from which the engine is being lifted.

13. Attach the overhead hoist to the engine. There are lift hooks provided on each engine.
14. Slowly and gently raise the engine through the hood area with the overhead hoist. Watch for any wiring they may have been overlooked in disconnecting which may include some optional accessory kit wiring. Some manoeuvring is required as the engine does not necessarily come straight up and out. Avoid damage to any other components as well as the upper body.
15. Place the engine on a clean work bench for service required.
16. Remove the engine from the engine mounting plate.





Installing The ODG 650/850 LX Engine

1. Install the engine mount plate to the engine block securing with four (4) HWHCS 5/16-18x2.00 GR8 YZ and four (4) HXFNN, 5/16-18 GR8 YZN. **Torque to 295 in-lbs (dry, non-lubricated condition).**
2. Slowly and gently lower the engine through the hood area. Some maneuvering is required as the engine does not necessarily drop straight in. Avoid damage to any vehicle components as well as the upper body.
3. Align the engine mounted frame holes with the holes in the lower main frame and reinstall each of the four (4) HHCS 3/8-16x2.0 GR5 BNZ & 108-06 Lock washers. Secure and torque each fastener to **31 FT-LBS** (Dry, Non-Lubricated Condition).
4. Reconnect engine connector plugs in the main wire harness located at the front of the engine.
5. Reconnect fuel vent from fitting on engine.
6. Reconnect the intake hose from the air filter cannister.

7. Reconnect the throttle cable at the engine cable retainer.
8. Reconnect the ground cable at the engine.
9. Reconnect the fuel line and return fuel line from the engine.
10. Locate the tailpipe assembly and reattach to the muffler with connecting springs.
11. Reinstall the heat deflector shield over the muffler.
12. Reconnect the battery positive (+) and negative (-) cables.

IMPORTANT

Reconnect the Positive (+) cable first, followed by the negative cable.